



Lesson 6: k-NN, Naive Bayes and Support Vector Machines

Technologies for Artificial Intelligence

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Before we start

- Exercise 5, which we discuss now
- The score that was too good, and what the honest number cost you

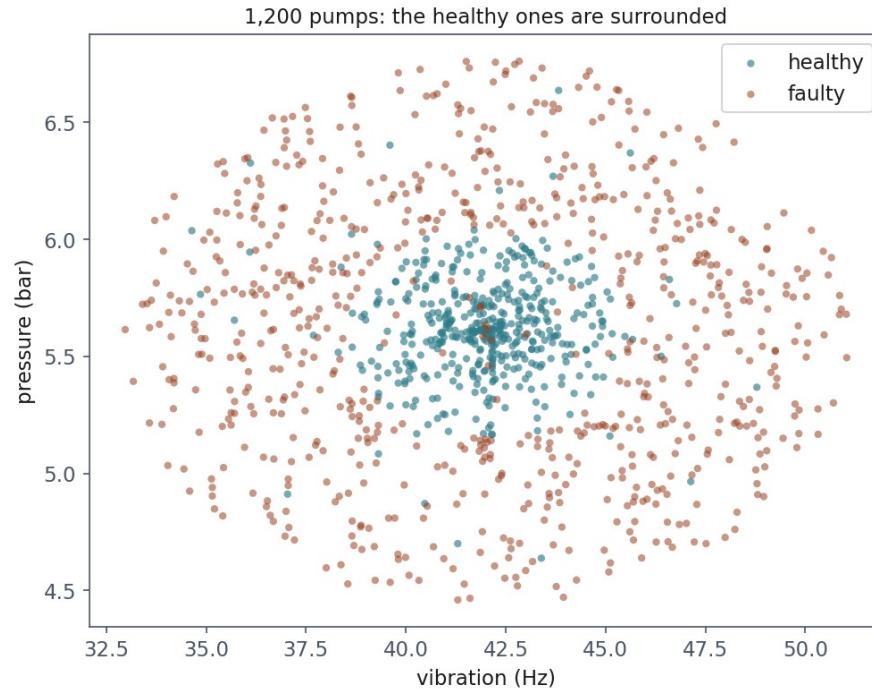
Today: the first lesson that offers a choice

- **k-nearest neighbours (k-NN)**, and the **curse of dimensionality** that limits it
- **Naive Bayes**: one strong assumption, a cheap model
- **Support vector machines (SVM)**: margins and kernels
- Which family to choose, and why that beats tuning

1,200 pumps, two readings each

- Vibration in Hz, pressure in bar
- A pump is faulty when its readings fall **outside the design envelope** (the range it was built to run in), too low as readily as too high
- **61.3%** faulty; 4% of the labels are deliberately flipped

The shape of the problem



What a straight boundary costs

Model	Cross-validated accuracy
Always predict the majority class	0.613
Logistic regression (lesson 4)	0.613 $\pm$ 0.000

k-nearest neighbours: the whole algorithm

- To classify a new point, find the **k closest** training points and take the majority vote
- There is no training step: “fitting” means **storing the data**
- Called a **lazy learner**: an unusually honest name for an algorithm
- In code:

```
KNeighborsClassifier(n_neighbors=5)
```

Distance is all it has

$$d(x, x') = \sqrt{\sum_{j=1}^n (x_j - x'_j)^2}$$

Scale first, or vibration decides everything

- Vibration runs over **tens of Hz**; pressure over **a couple of bar**
- Unscaled, a one-bar pressure difference is invisible beside a ten-Hz one
- Lesson 2's scaling argument, arriving with **immediate consequences**
- In code: `make_pipeline(StandardScaler(), KNeighborsClassifier(5))`

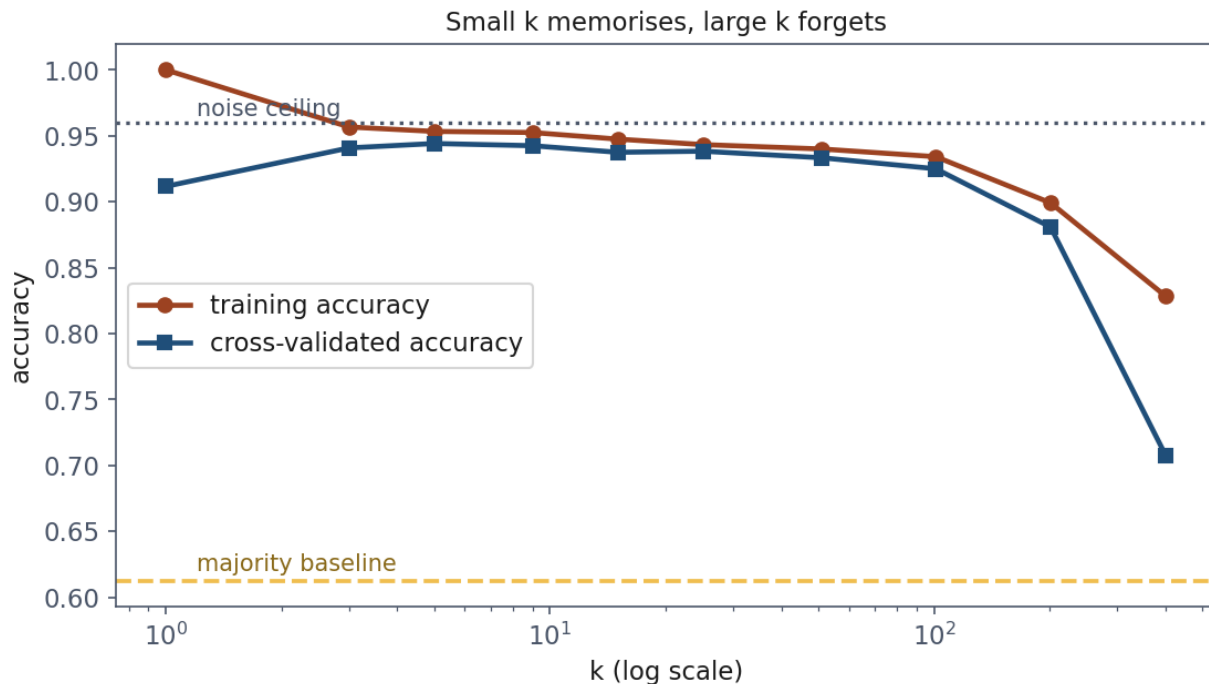
k is the bias-variance dial, made visible

- **Small k**: the boundary follows every point, mislabelled ones included. Low bias, high variance
- **Large k**: the vote is taken over a wide neighbourhood, and eventually stops following real structure. High bias, low variance
- At $k = 1$ the training accuracy is **exactly 1.000**, unless two identical points carry different labels

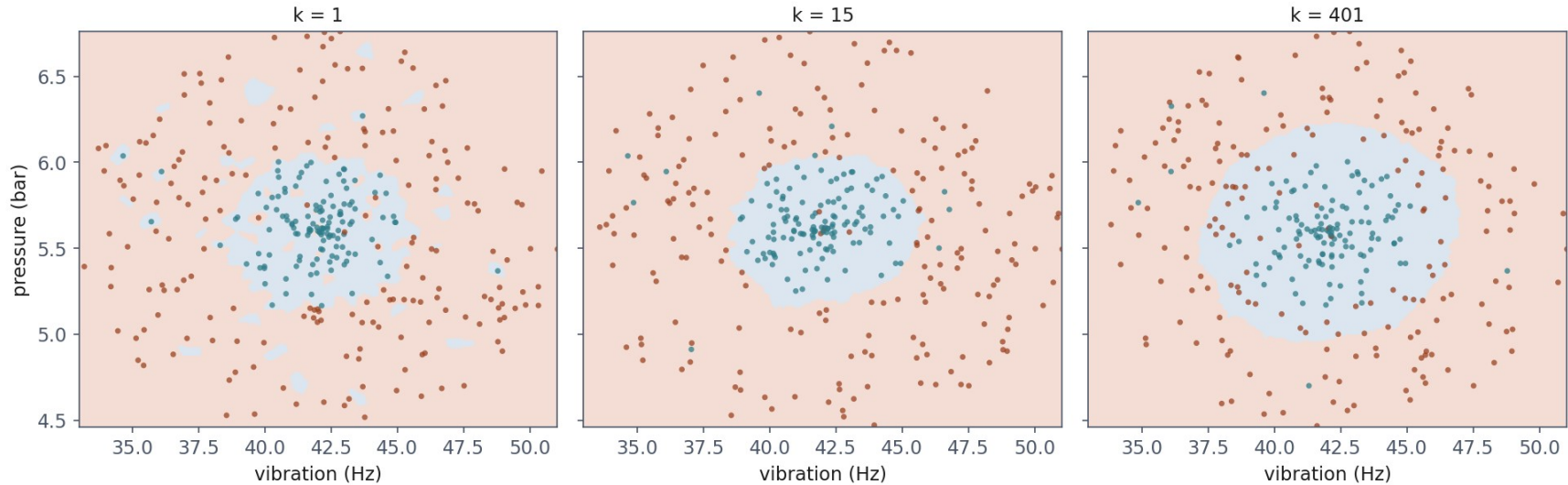
Measured on the pumps

k	Training	Cross-validated
1	1.000	0.912
5	0.953	0.944
15	0.948	0.938
51	0.940	0.933
401	0.828	0.708

The two curves, and the two lines that bound them



The same data at three values of k



What k-NN costs

- No training time at all: you pay at **every prediction** instead
- A naive implementation compares the query against **every** training row: $O(mn)$ for m rows and n features
- **The model is the dataset.** You cannot ship one without the other

Now the price: add columns of nothing

- Extra columns of **pure noise**, drawn from a normal distribution
- The two real readings are **untouched**: the problem is exactly as solvable
- Only the number of columns changes

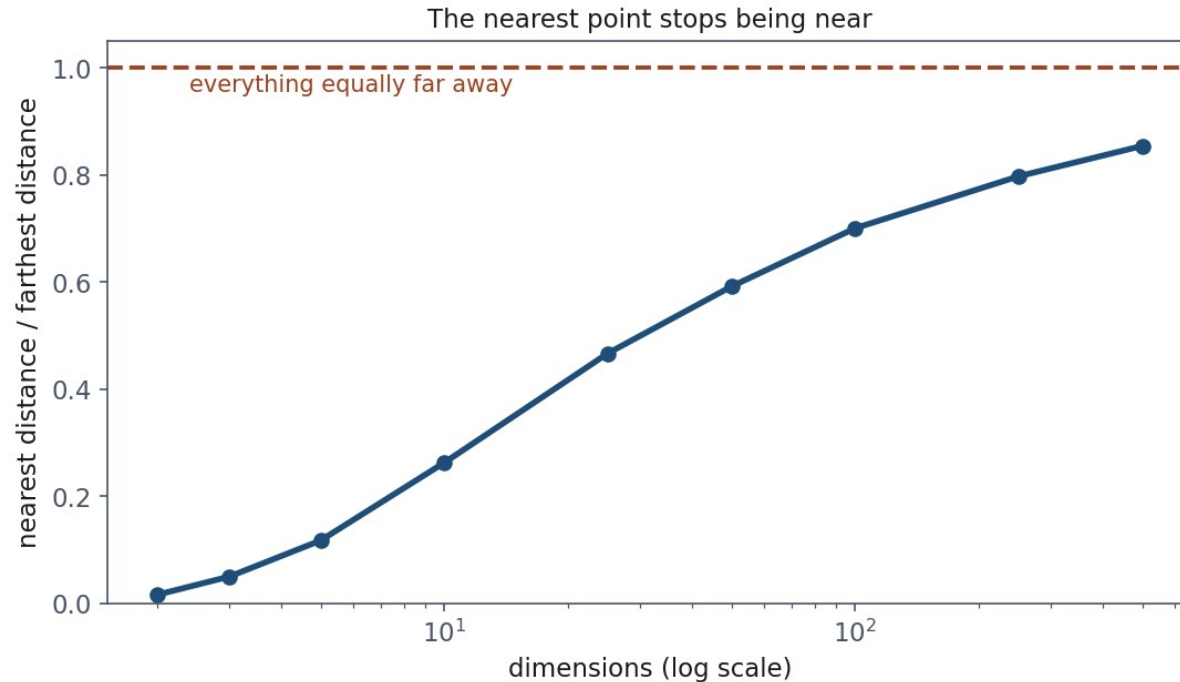
The signal never left

Total columns	Accuracy	Above baseline
2	0.938	+0.325
12	0.762	+0.149
27	0.662	+0.049
52	0.602	-0.011
102	0.578	-0.035

Why: distances stop varying

- Scatter points in a cube, pick one, measure its **nearest** and its **farthest** neighbour
- In two dimensions those are very different numbers, which is what makes “nearest” a meaningful word
- Each new dimension adds its own squared difference, and the contributions **average out**

The nearest point stops being near



Nearest divided by farthest

Dimensions	nearest ÷ farthest
2	0.016
10	0.263
50	0.592
100	0.701
500	0.855

In 100 dimensions the nearest point is 70% as far as the farthest

- In **two** dimensions it is **2%**: “nearest” picks out something special
- In **one hundred** it is **70%**: the nearest point is barely nearer than a random one
- A vote among “the five nearest” becomes a vote among five taken at random

What the curse is, and is not

- **Not** that high-dimensional problems are unlearnable: lesson 9's networks work in thousands of dimensions
- That **methods built on distance lose their footing**, because the quantity they depend on stops varying
- k-NN, k-means (lesson 8), and the radial basis function (RBF) kernel of support vector machines, later today

Notebook 1, live

section, and its slides	you call (→ the number to come back with)
the straight line, 4-6	<code>LogisticRegression</code> scored by <code>cross_val_score</code> over <code>StratifiedKFold</code> folds → 0.613
k-NN, scaled first, 7-9	<code>make_pipeline(StandardScaler(), KNeighborsClassifier(k))</code>
choosing k, 10-13	<code>cross_val_score</code> for each k → 0.944 at k = 5
columns of noise, 15-16	the same pipeline on <code>load_with_noise(d)</code> → 0.602 at 52 columns
distances stop varying, 17-20	NumPy only: nearest ÷ farthest → 0.701 in 100 dimensions

Break

- Twelve minutes

Naive Bayes: turn the question around

- We want the probability of the class **given** the readings
- The prior $P(y = c)$ is the class's share of the data; the denominator is the same for every class and cancels
- Everything hard sits in one term: the probability of **this exact combination of readings** among examples of that class

Bayes' rule, applied to a class

$$P(y = c \mid x) = \frac{P(x \mid y = c) P(y = c)}{P(x)}$$

The decision drops the denominator

$$\hat{y} = \arg \max_c P(x \mid y = c) P(y = c)$$

The assumption: independent, given the class

- **Given the class, the features are independent of one another**
- The joint density then factorises into n one-dimensional densities
- Each factor is estimated from the rows of that class alone
- In code: `GaussianNB()`

The factorisation the assumption
buys

$$P(x \mid y = c) = \prod_{j=1}^n P(x_j \mid y = c)$$

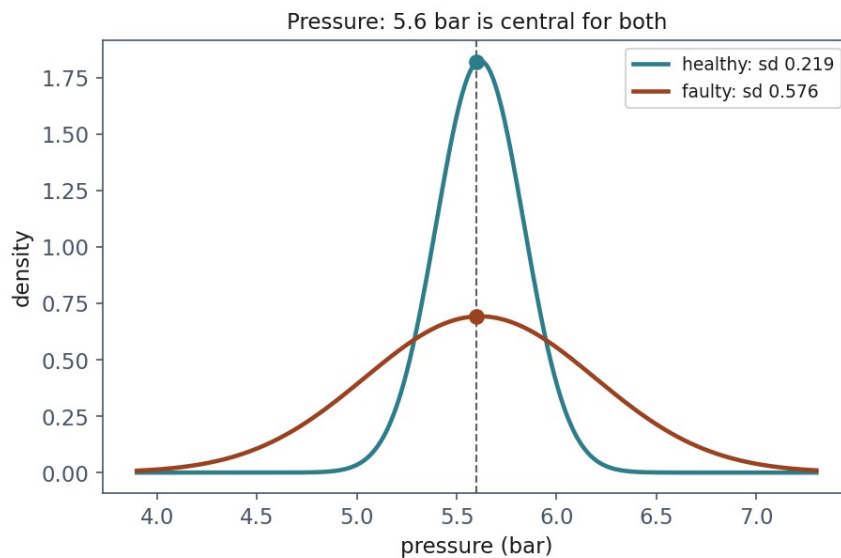
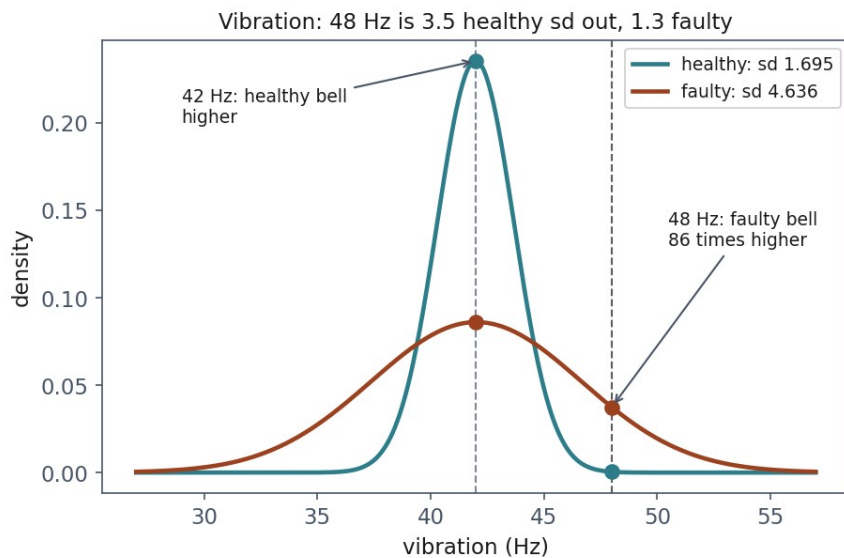
Gaussian: each factor is a bell curve

$$P(x_j \mid y = c) = \frac{1}{\sqrt{2\pi\sigma_{jc}^2}} \exp \left[-\frac{(x_j - \mu_{jc})^2}{2\sigma_{jc}^2} \right]$$

Ten numbers, and the widths decide

Class	Prior	Vibration mean	Vibration sd	Pressure mean	Pressure sd
Healthy	0.3875	41.986 Hz	1.695	5.617 bar	0.219
Faulty	0.6125	42.015 Hz	4.636	5.616 bar	0.576

48 Hz: one bell is 86 times the other



Add the natural logs: same winner

$$\ln \left[P(y = c) \prod_j P(x_j \mid y = c) \right] = \ln P(y = c) + \sum_j \ln P(x_j \mid y = c)$$

48 Hz, 5.6 bar: scores and their logs

Class		Prior	Vibration density	Pressure density	Product, or sum of logs
Healthy	value	0.3875	0.000433	1.818	0.000305
Healthy	ln	-0.948	-7.745	+0.598	-8.095
Faulty	value	0.6125	0.0374	0.692	0.01585
Faulty	ln	-0.490	-3.286	-0.368	-4.144

From the two totals to probabilities

Step	Healthy	Faulty
1. Log total, previous slide	-8.095	-4.144
2. Undo the log: $e^{\text{total}} = \text{prior} \times \text{densities}$	0.000305	0.01585
3. Divide by their sum: $0.01616 = P(x)$, by total probability	0.019	0.981

Why the gap is enough

- Divide by the faulty score: $1 / (1 + \text{healthy score} \div \text{faulty score})$
- $e^a \div e^b = e^{(a - b)}$, so the ratio is $e^{(-3.951)} = 0.0192$
- $P(\text{faulty}) = 1 / 1.0192 = \mathbf{0.981}$; $P(\text{healthy}) = 1 - 0.981 = \mathbf{0.019}$
- Odds: $0.01585 \div 0.000305 \approx \mathbf{52 \text{ to } 1}$
- With many features e^{total} rounds to 0 in the computer; the gap does not

A wonderful bargain, and almost never true

- Training is **a single pass**; adding features costs almost nothing
- But vibration and pressure are both driven by the operating point
- “New” is not independent of “York” given the topic. Symptoms co-occur
- So the question is not whether it holds: it is **when being wrong about it costs you nothing**

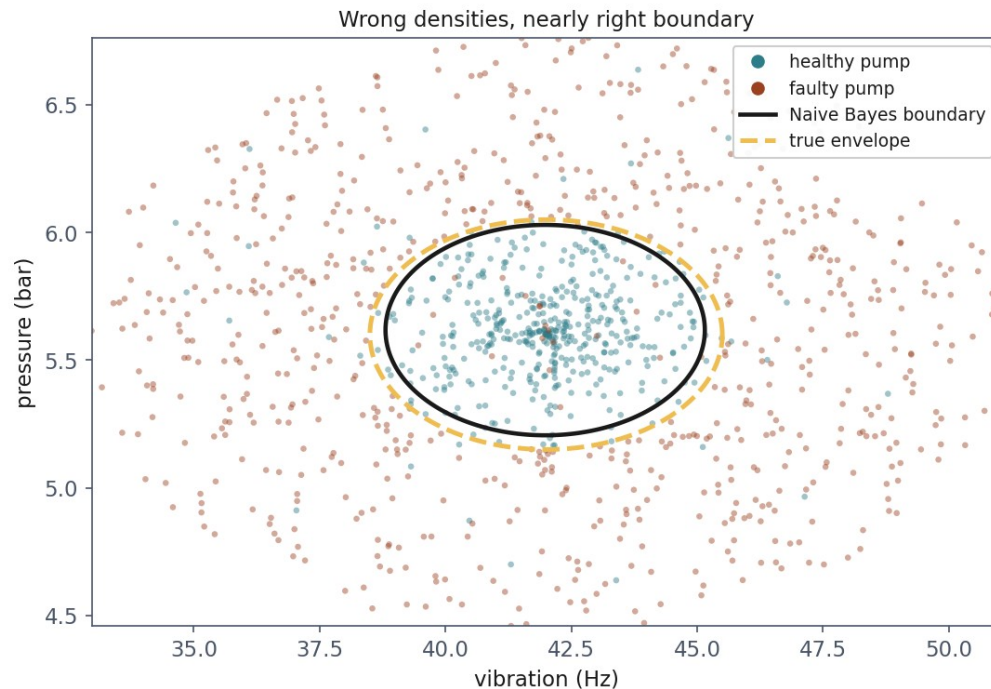
On the pumps, it scores 0.933

Model	Accuracy
Majority baseline	0.613
Logistic regression	0.613
Gaussian Naive Bayes	0.933
k-NN, $k = 5$	0.944

Uncorrelated, but not independent

Correlation, within the class	vibration and pressure	their squared distances from the design point
healthy pumps	-0.006	+0.168
faulty pumps	-0.049	-0.420

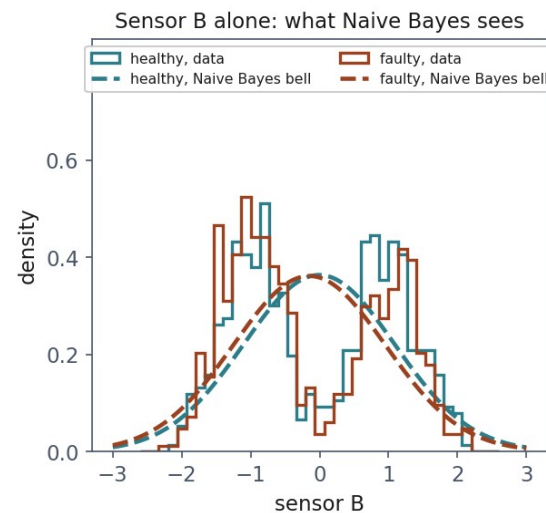
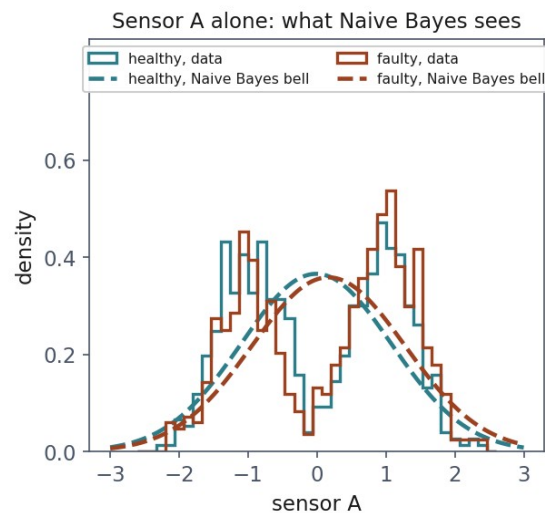
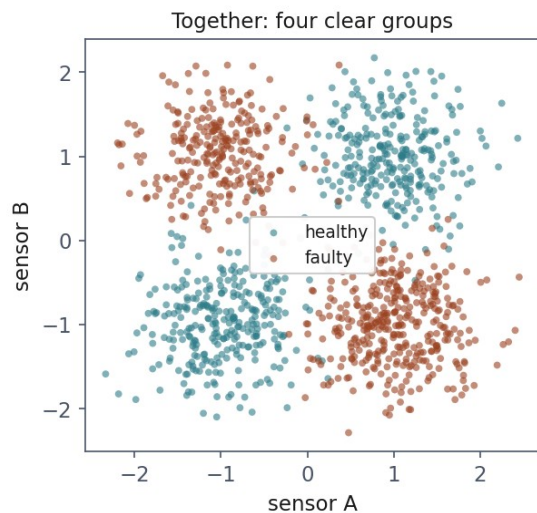
Wrong densities, right boundary



One step away: when the signal is an interaction

- A second pair of sensors on the same fleet
- Faulty when **exactly one** of the two readings is high
- Both high is the designed high-load mode; both low is idle; one without the other is a mismatch between demand and delivery

What Naive Bayes gets to see



0.404: below the baseline, and below chance

Model	Accuracy
Majority baseline	0.523
Gaussian Naive Bayes	0.404
Logistic regression	0.393
SVM, linear kernel	0.606
k-NN, k = 5	0.967
SVM, RBF kernel	0.972

Its probabilities are not probabilities

- Confidence, the probability given to the predicted class: **0.567** when correct, **0.555** when wrong
- It cannot tell the difference between the two situations
- With features that repeat each other it fails the other way: 0.999 reported, with an accuracy nothing like that
- **The ranking may be useful while the probabilities are not**

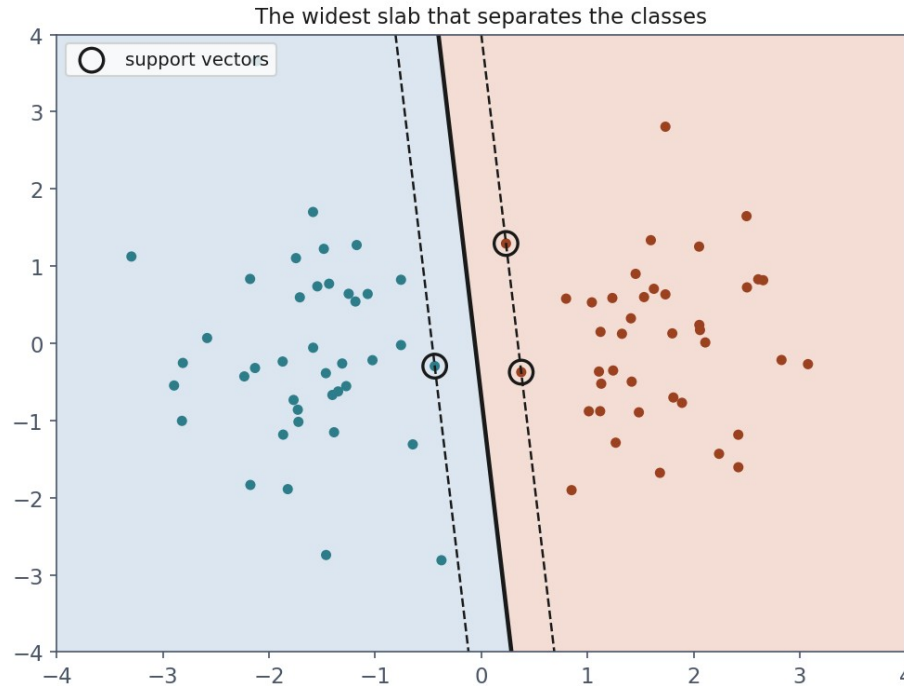
Notebook 2, live

section, and its slides	you call (→ the number to come back with)
on the pumps, 37	GaussianNB beside KNeighborsClassifier and LogisticRegression → 0.933
what training computed, 29-35	.theta_, .var_, .class_prior_, predict_proba → 0.981 at 48 Hz
checking the assumption, 38-39	X[y == c].corr(), within each class → −0.420
the interacting sensors, 40-42	the same models, plus svc → 0.404
confidence, right and wrong, 43	train_test_split, then predict_proba → 0.567 against 0.555

The margin: a criterion that is not the error

- On separable data there are **infinitely many** separating lines, and every one has **zero training error**
- Minimising the error therefore cannot choose between them
- The SVM picks the widest **slab**: push it out until it touches the nearest point of each class
- A boundary passing close to a point is one small perturbation from getting it wrong

Three points decide everything



Not separable? Charge for the violations

- Our labels are 4% flipped, so the strict problem has **no solution at all**
- Let each point violate the margin by ξ_i , and charge C for every unit of it
- $\xi_i = 0$ outside the slab, between 0 and 1 inside it, above 1 on the wrong side
- That is the **hinge loss**, $\max(0, 1 - y \cdot f(x))$ with labels ± 1 : zero for a point with room to spare

The hinge stops at zero



A wide slab, minus what the violations cost

$$\min_{w, b, \xi} \frac{1}{2} \|w\|^2 + C \sum_i \xi_i$$

C is the price of a training error

- **Large C**: violations are expensive, so the model accepts a narrower margin to classify more training points. Low bias, high variance
- **Small C**: a wider, calmer boundary, at the cost of some errors
- C plays the role that k played in k -NN and λ in lesson 3
- Note the direction: **large C means less regularisation**

The best straight line is still a straight line

Model	Accuracy	Support vectors
SVM, linear kernel	0.613 $\pm$ 0.000	947 of 1,200 (79%)
SVM, RBF kernel	0.947 $\pm$ 0.005	278 of 1,200 (23%)

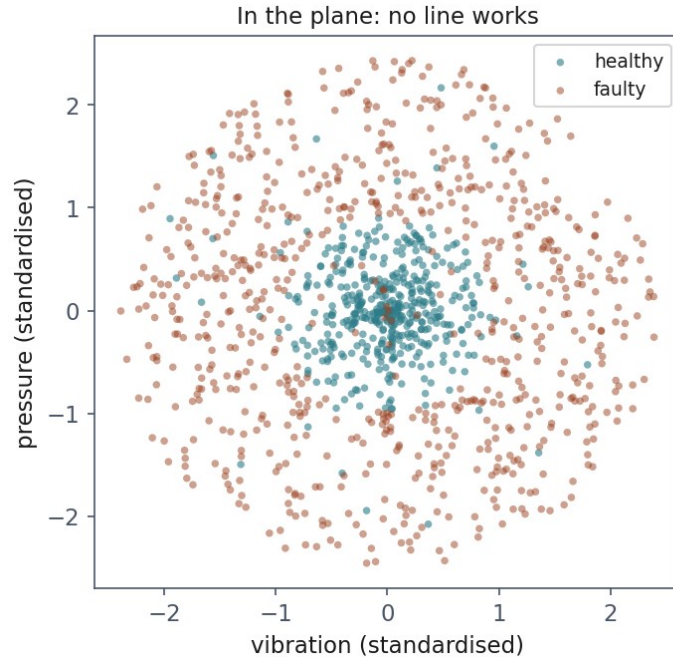
A high support-vector fraction is a free warning

- The linear model, `SVC(kernel="linear")`, needs **79%** of the training set
- The RBF model needs **23%**
- Almost every point sits on or inside the margin: there is no slab that separates anything
- The count comes free with `fit`, as `len(model.support_)`: look at it

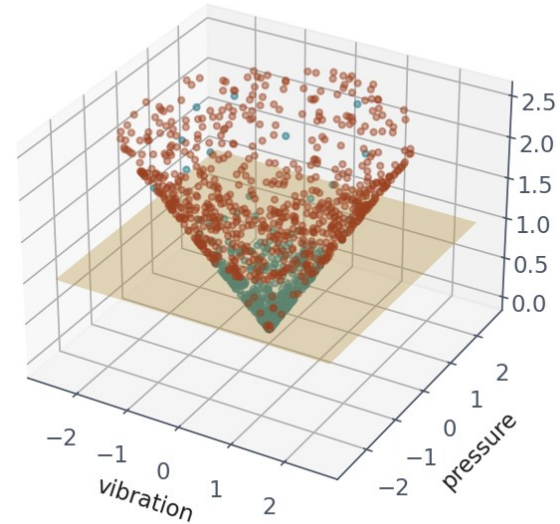
The kernel trick: they needed different coordinates

- Healthy pumps in a disc, faulty ones around it: no line works **in the plane**
- Add a third coordinate: the **distance from the design point**
- Healthy pumps rise a little, faulty ones a lot, and a flat plane separates them perfectly
- The classes were always separable. They were measured in the wrong coordinates

The same pumps, lifted



Lifted: a flat plane does
height = distance from the design point



And the map is never computed

- The solution depends on the data **only through inner products** between pairs of points
- So we need a function returning the inner product **in the new space**, while computing only with the original one
- The radial basis function is scikit-learn's default: one exponential per pair, for an **infinite-dimensional** space

The boundary is a weighted vote of
the pumps

$$w = \sum_{i=1}^m a_i y_i x_i, \quad a_i \geq 0$$

The kernel trick by hand: $25 = 25$

Route	Computation	Result
Lift, then multiply	$(1, 2.83, 4) \cdot (9, 4.24, 1) = 9 + 12 + 4$	25
Never lift	$((1, 2) \cdot (3, 1))^2 = 5^2$	25

The radial basis function, written out

$$K(x, x') = \exp(-\gamma \|x - x'\|^2)$$

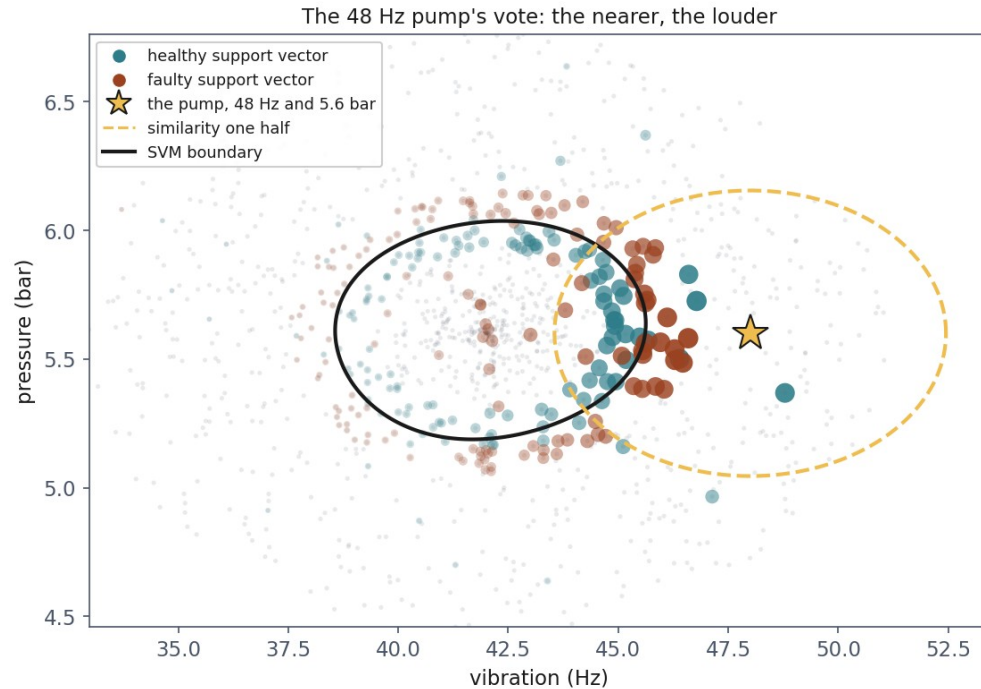
Training: one weight per pump

- Weights adjusted **two at a time**; no learning rate, same answer every run
- **922** pumps: weight 0, forgotten
- **268** at the ceiling $C = 1$, **10** on the edge: the 278 support vectors
- The model: 278 points, their weights, and $b = +1.105$

The SVM on 48 Hz, 5.6 bar: +1.883

Term	Value
votes from faulty support vectors	+37.455
votes from healthy support vectors	-36.677
intercept b	+1.105
total	+1.883

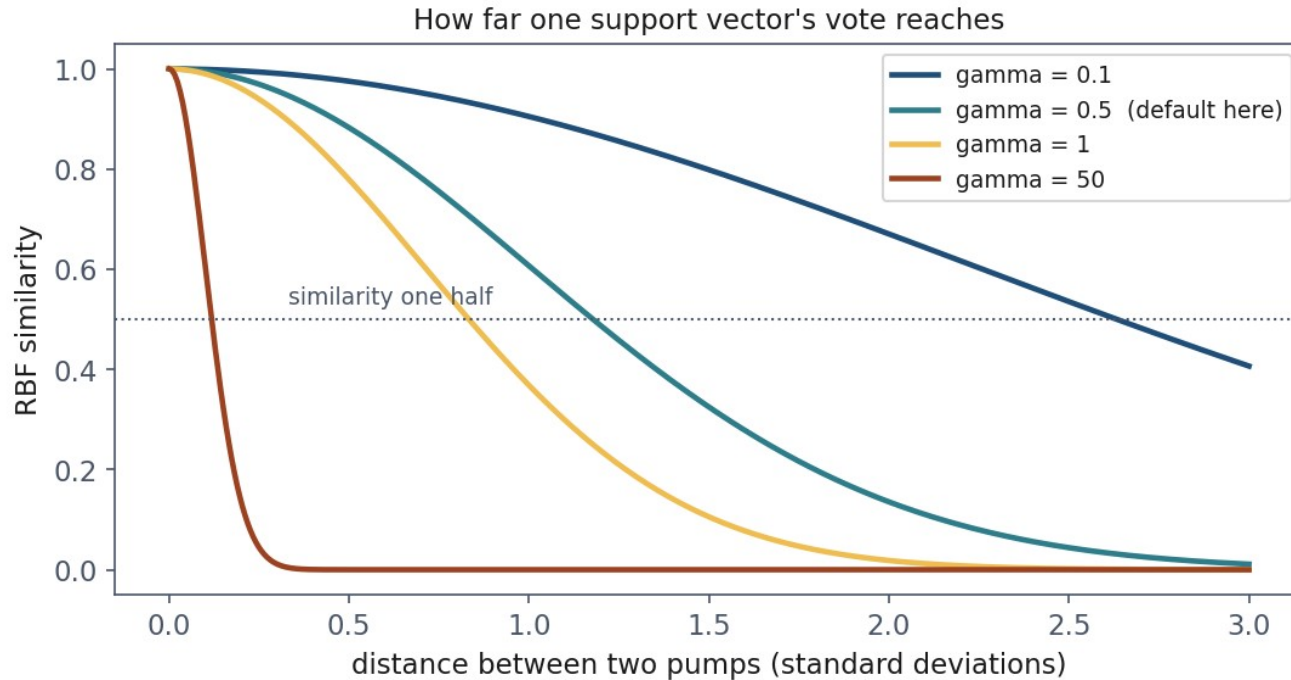
One pump's vote, drawn



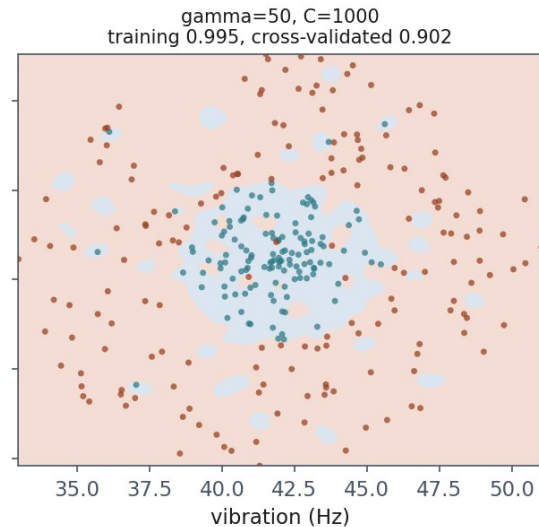
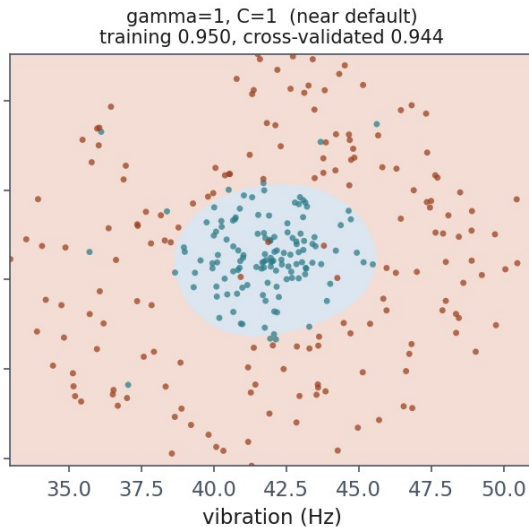
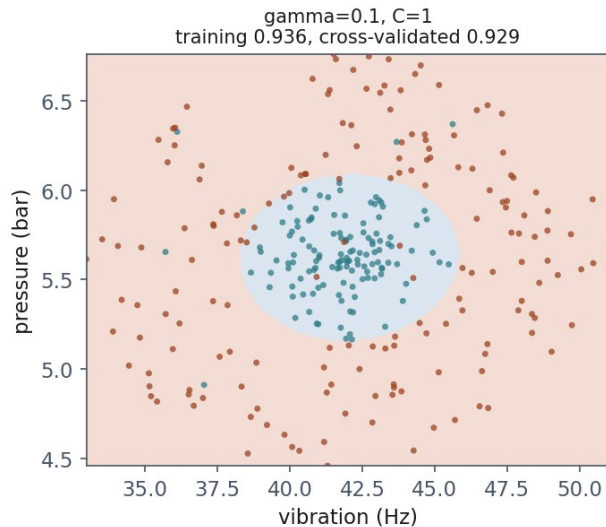
The highest training score, the lowest honest one

Setting	Training	Cross-validated
$\gamma = 0.1, C = 1$	0.936	0.929
$\gamma = 1, C = 1$	0.950	0.944
$\gamma = 50, C = 1000$	0.995	0.902

How far one vote reaches



Overfitting you can see



Notebook 3, live

section, and its slides	you call (→ the number to come back with)
the widest slab, 45-48	<code>SVC(kernel="linear"), .n_support_, decision_function</code> → 3 of 80 points
a straight line still fails, 51-52	<code>cross_val_score</code> , linear against RBF kernel → 947 against 278 support vectors
the kernel trick, 53-58	NumPy only: $(a \cdot b)^2$ against the lifted product → $25 = 25$
training and one prediction, 59-61	<code>.support_, .dual_coef_, .intercept_</code> → +1.883
gamma and C, 62-64	<code>SVC(gamma=..., C=...)</code> in the pipeline → 0.944 at $\gamma = 1$

The three compared, on identical data

Model	Accuracy
Majority baseline	0.613
Logistic regression (lesson 4)	0.613
SVM, linear kernel	0.613
Gaussian Naive Bayes	0.933
k-NN, $k = 5$	0.944
SVM, RBF kernel	0.947
<i>noise ceiling</i>	≈ 0.96

How to choose, in practice

Situation	Reach for
Few features, plenty of data, odd-shaped boundary	k-NN, or an RBF SVM
Very many features, little data	Naive Bayes
Need a calibrated probability	Not Naive Bayes
Model must be small, or fast at prediction	SVM, not k-NN
The signal is an interaction between features	k-NN or a kernel

What to take away

- **No line separates the pumps:** 0.613, the base rate
- **k is the bias-variance dial**, made visible
- **In 100 dimensions the nearest point is 70% as far as the farthest**
- **Naive Bayes:** 0.933 where its false assumption is cheap, **0.404** where not
- **Choosing the family beats tuning the wrong one:** 0.613 to 0.947

Homework

- **Exercise 6**, discussed at the start of **Friday 13 November**
- A dataset, three families, and a defence of which one you chose